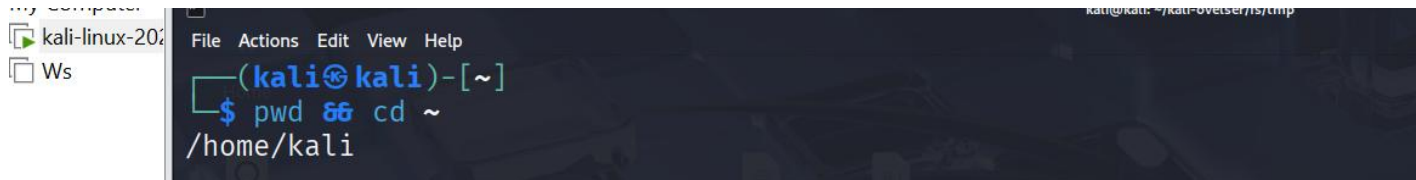


En rundtur i Linux - Øvelser i terminal

1) Filsystem

1. Find din nuværende sti og gå til din hjemmemappe.



```
(kali㉿kali)-[~]  
$ pwd  
/home/kali  
$ cd ~
```

2. Opret ~/kali-ovelses/fs med data og tmp som undermapper.



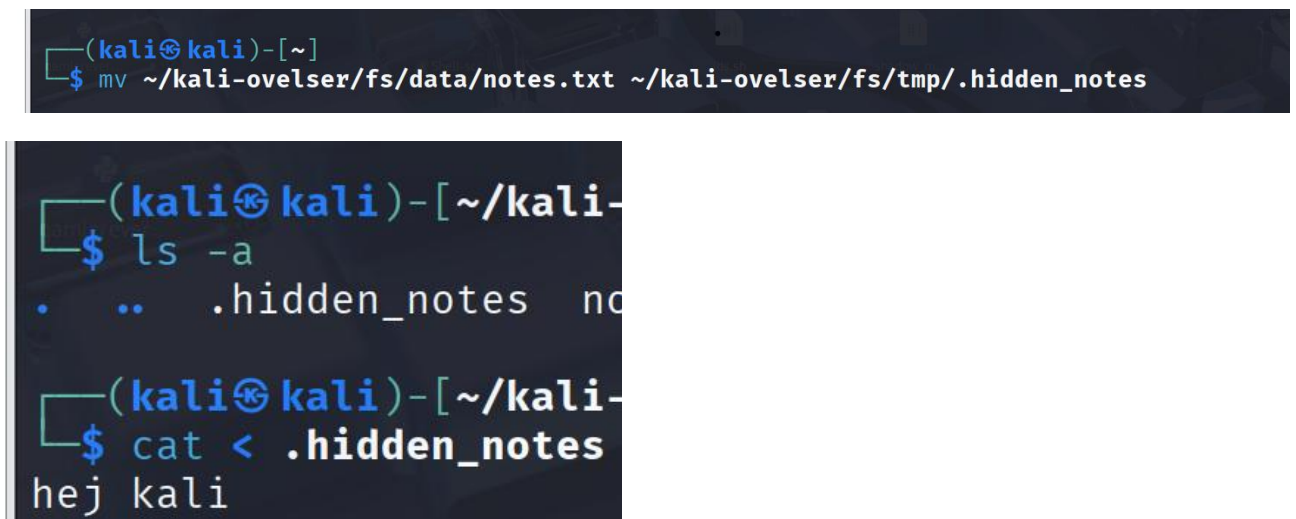
```
(kali㉿kali)-[~]  
$ mkdir -p ~/kali-ovelses/fs/{data,tmp}
```

3. Lav filen notes.txt i data med teksten hej kali.



```
(kali㉿kali)-[~]  
$ echo "hej kali" > ~/kali-ovelses/fs/data/notes.txt
```

4. Flyt notes.txt til tmp og omdøb den til .hidden_notes.



```
(kali㉿kali)-[~]  
$ mv ~/kali-ovelses/fs/data/notes.txt ~/kali-ovelses/fs/tmp/.hidden_notes
```



```
(kali㉿kali)-[~/kali-ovelses/fs/tmp]  
$ ls -la  
.  ..  .hidden_notes  
$ cat < .hidden_notes  
hej kali
```

2) Brugere og grupper

1. Vis dit brugernavn og hvilke grupper du er i.

```
(kali㉿kali)-[~]  
$ id  
uid=1000(kali) gid=1000(kali) groups=1000(kali),25(floppy),27(sudo),29(audio),30(dip),44(video),103(scanner),107(bluetooth),125(lpadmin),1001(lab)
```

2. Slå din bruger op i /etc/passwd.

```
(kali㉿kali)-[~]  
$ grep "^$USER:" /etc/passwd  
kali:x:1000:1000::/home/kali:/usr/bin/zsh
```

3. (Hvis muligt) Opret gruppen lab og tilføj din bruger til den.

```
(kali㉿kali)-[~]  
$ sudo groupadd lab 2>/dev/null || true && sudo usermod -aG lab $USER  
[sudo] password for kali:
```

3) Processer

1. Vis processer for din bruger.

```
(kali㉿kali)-[~]  
$ ps -u $USER  
  
  PID TTY          TIME CMD  
 1220 ?            00:00:00 systemd  
 1222 ?            00:00:00 (sd-pam)  
 1242 ?            00:00:00 dbus-daemon  
 1243 ?            00:00:00 pipewire  
 1244 ?            00:00:00 pipewire  
 1246 ?            00:00:00 wireplumber  
 1247 ?            00:00:00 pipewire-pulse  
 1248 ?            00:00:00 gnome-keyring-d
```

2. Find PID for din nuværende shell.

```
(kali㉿kali)-[~]  
$ echo $$  
  
18339
```

3. Start sleep 60 i baggrunden og vis at den kører.

```
(kali㉿kali)-[~]  
$ sleep 60 & jobs  
[1] 21368  
[1] + running      sleep 60
```



4) Resurser (CPU, RAM, disk)

1. Vis et snapshot af CPU og RAM.

```
(kali㉿kali)-[~]  
$ top -b -n1 | head -n 10  
top - 13:46:08 up 41 min, 1 user, load average: 0.07, 0.13, 0.15  
Tasks: 217 total, 2 running, 215 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 3.8 us, 5.7 sy, 0.0 ni, 90.6 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st  
MiB Mem : 3912.8 total, 2486.6 free, 929.5 used, 758.5 buff/cache  
MiB Swap: 953.7 total, 953.7 free, 0.0 used. 2983.2 avail Mem  
  
    PID USER      PR  NI   VIRT   RES    SHR S  %CPU  %MEM     TIME+ COMMAND  
    984 root        20   0 423000 140300 65984 S   7.7   3.5   1:52.51 Xorg  
   22304 kali       20   0  10420   5344   3296 R   7.7   0.1   0:00.04 top  
        1 root        20   0  23896  14520  10444 S   0.0   0.4   0:03.24 systemd
```

2. Vis brug af monterede filerystemer.

```
(kali㉿kali)-[~]  
$ df -h  
  
Filesystem      Size  Used Avail Use% Mounted on  
udev            1.9G   0    1.9G   0% /dev  
tmpfs           392M  1.2M  391M   1% /run  
/dev/sda1       79G   19G   56G  26% /  
tmpfs           2.0G  4.0K  2.0G   1% /dev/shm  
tmpfs           5.0M   0    5.0M   0% /run/lock  
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-journald.service  
tmpfs           2.0G  88K  2.0G   1% /tmp  
tmpfs           1.0M   0    1.0M   0% /run/credentials/getty@tty1.service  
tmpfs           392M  128K  392M   1% /run/user/1000
```

3. Mål hvor lang tid ls / tager.

```
(kali㉿kali)-[~]  
$ time ls / >/dev/null  
  
real    0.01s  
user    0.00s  
sys     0.01s  
cpu     92%
```

5) Netværk

1. Vis dine netværksinterfaces og IP-adresser.

```
(kali㉿kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:37:48:30 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.231.128/24 brd 192.168.231.255 scope global dynamic noprefixroute eth0  
        valid_lft 971sec preferred_lft 971sec  
    inet6 fe80::c571:9547:b4b0:7061/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever
```

2. Ping kali.org med 3 pakker.

```
(kali㉿kali)-[~]  
$ ping -c 3 kali.org  
  
PING kali.org (104.18.5.159) 56(84) bytes of data:  
64 bytes from 104.18.5.159: icmp_seq=1 ttl=128 time=29.1 ms  
64 bytes from 104.18.5.159: icmp_seq=2 ttl=128 time=16.3 ms  
64 bytes from 104.18.5.159: icmp_seq=3 ttl=128 time=15.7 ms  
  
— kali.org ping statistics —  
3 packets transmitted, 3 received, 0% packet loss, time 2001ms  
rtt min/avg/max/mdev = 15.687/20.366/29.085/6.170 ms
```

3. Se hvilke processer der lytter på lokale porte.

```
(kali㉿kali)-[~]  
$ sudo ss -tulpn  
Netid    State    Recv-Q    Send-Q
```

🔧 6) Systeminfo & environment

1. Vis kernel-version og maskine-arkitektur.

```
(kali㉿kali)-[~]  
$ uname -r && uname -m  
  
6.12.25-amd64  
x86_64
```

2. Vis miljøvariablen PATH.

```
(kali㉿kali)-[~]  
$ echo "$PATH"  
  
/home/kali/.local/bin:/usr/local/sbin:/usr/sbin:/sbin:/usr/local/bin:/usr/bin:/bin:/usr/  
local/games:/usr/games:/home/kali/.dotnet/tools
```

📦 7) Installering & opdatering (APT)

1. Opdater pakkelister.

```
(kali㉿kali)-[~]  
$ sudo apt update  
  
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]  
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [20.9 MB]  
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.6 kB]  
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [114 kB]  
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [25.6 kB]  
Get:6 http://kali.download/kali kali-rolling/contrib amd64 Packages [100 kB]
```

2. Søg efter pakken jq.

```
(kali㉿kali)-[~]  
$ apt search jq | head -n 10  
  
WARNING: apt does not have a stable CLI interface.  
  
Sorting ...  
Full Text Search ...  
dasel/kali-rolling 2.8.1-1 amd64  
  Query, update and convert data structures from the  
  
fq/kali-rolling 0.9.0-2 amd64  
  jq for binary formats (program)  
  
gojq/kali-rolling 0.12.13-1 amd64  
  pure Go implementation of jq (program)
```


3. Installer jq, vis versionen, og fjern den igen.

```
(kali@kali)-[~]
$ sudo apt install -y jq && jq --version && sudo apt remove -y jq

Upgrading:
  libjq1

Installing:
  jq

Summary:
  Upgrading: 1, Installing: 1, Removing: 0, Not Upgrading: 1630
  Download size: 252 kB
  Space needed: 136 kB / 59.7 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 libjq1 amd64 1.8.1-4 [167 kB]
Get:2 http://mirrors.dotsrc.org/kali kali-rolling/main amd64 jq amd64 1.8.1-4 [85.4 kB]
```

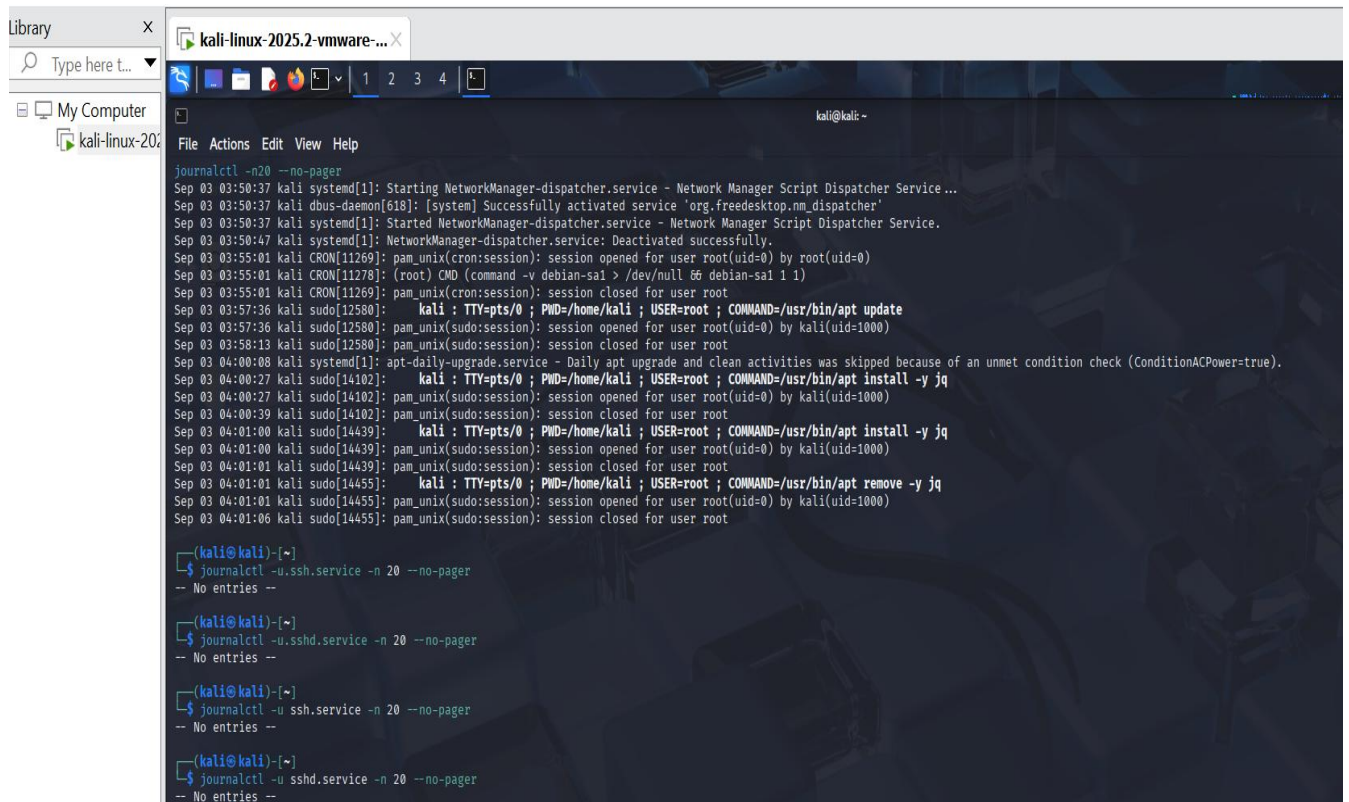
```
REMOVING:
  jq

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 1630
  Freed space: 136 kB

(Reading database ... 417169 files and directories currently installed.)
Removing jq (1.8.1-4) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.2.7) ...
```

8) Logging (basic)

1. Se de sidste 20 linjer i systemjournalen.
2. Se de sidste 20 linjer for ssh-servicen.



The screenshot shows a Kali Linux terminal window with the following content:

```
File Actions Edit View Help

journalctl -n20 --no-pager
Sep 03 03:50:37 kali systemd[1]: Starting NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service ...
Sep 03 03:50:37 kali dbus-daemon[618]: [system] Successfully activated service 'org.freedesktop.nm_dispatcher'
Sep 03 03:50:37 kali systemd[1]: Started NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service.
Sep 03 03:50:47 kali systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
Sep 03 03:55:01 kali CRON[11269]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Sep 03 03:55:01 kali CRON[11278]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
Sep 03 03:55:01 kali CRON[11269]: pam_unix(cron:session): session closed for user root
Sep 03 03:57:36 kali sudo[12580]:      kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/apt update
Sep 03 03:57:36 kali sudo[12580]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Sep 03 03:58:13 kali sudo[12580]: pam_unix(sudo:session): session closed for user root
Sep 03 04:00:08 kali systemd[1]: apt-daily-upgrade.service - Daily apt upgrade and clean activities was skipped because of an unmet condition check (ConditionACPower=true).
Sep 03 04:00:27 kali sudo[14102]:      kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/apt install -y jq
Sep 03 04:00:27 kali sudo[14102]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Sep 03 04:00:39 kali sudo[14102]: pam_unix(sudo:session): session closed for user root
Sep 03 04:01:00 kali sudo[14439]:      kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/apt install -y jq
Sep 03 04:01:00 kali sudo[14439]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Sep 03 04:01:01 kali sudo[14439]: pam_unix(sudo:session): session closed for user root
Sep 03 04:01:01 kali sudo[14455]:      kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/apt remove -y jq
Sep 03 04:01:01 kali sudo[14455]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Sep 03 04:01:06 kali sudo[14455]: pam_unix(sudo:session): session closed for user root

(kali@kali)-[~]
$ journalctl -u.ssh.service -n 20 --no-pager
-- No entries --

(kali@kali)-[~]
$ journalctl -u.sshd.service -n 20 --no-pager
-- No entries --

(kali@kali)-[~]
$ journalctl -u ssh.service -n 20 --no-pager
-- No entries --

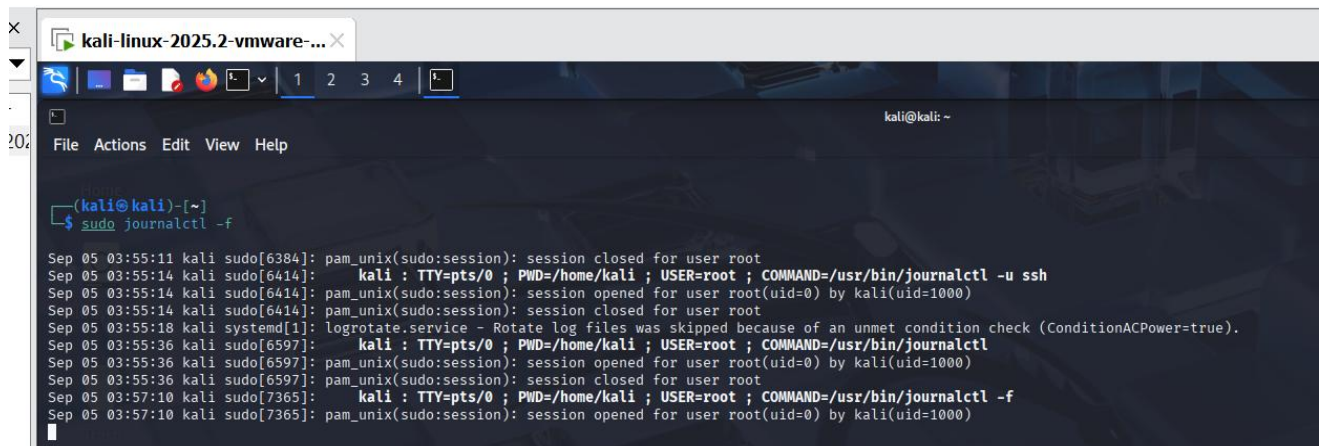
(kali@kali)-[~]
$ journalctl -u sshd.service -n 20 --no-pager
-- No entries --
```

3. Se de seneste APT-hændelser (pakkehistorik).

```
(kali@kali)-[~]
$ grep -E '^(\Start-Date|CommandLine):' /var/log/apt/history.log | tail -n 20

Start-Date: 2025-05-29 19:03:50
CommandLine: apt-get install -y ?priority(standard) !?section(lib) !(~napt-listchanges|~ndebian-faq|~ndoc-debian|~npython3-reportbug|~nreportbug|~nwamerican)
Start-Date: 2025-05-29 15:04:45
CommandLine: apt-get -y -o=quiet::NoUpdate=1 install kali-linux-core kali-linux-firmware kali-desktop-xfce
Start-Date: 2025-05-29 15:11:55
```

4. Følg i realtid en logfil i ~10 sekunder og stop med Ctrl+C.



```
kali-linux-2025.2-vmware-...
File Actions Edit View Help

(kali@kali)-[~]
$ sudo journalctl -f

Sep 05 03:55:11 kali sudo[6384]: pam_unix(sudo:session): session closed for user root
Sep 05 03:55:14 kali sudo[6414]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl -u ssh
Sep 05 03:55:14 kali sudo[6414]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Sep 05 03:55:14 kali sudo[6414]: pam_unix(sudo:session): session closed for user root
Sep 05 03:55:18 kali systemd[1]: logrotate.service - Rotate log files was skipped because of an unmet condition check (ConditionACPower=true).
Sep 05 03:55:36 kali sudo[6597]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl
Sep 05 03:55:36 kali sudo[6597]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Sep 05 03:55:36 kali sudo[6597]: pam_unix(sudo:session): session closed for user root
Sep 05 03:57:10 kali sudo[7365]:    kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl -f
Sep 05 03:57:10 kali sudo[7365]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
```

disse kommandoer virkede ikke - **sudo tail -f /var/log/auth.log** og **/var/log/secure**) alternativt brugte jeg **sudo journalctl -f** som også følger systemlogge i realtid.

5. List de 5 største filer i /var/log (overblik).

```
(kali@kali)-[~]
$ sudo ls -lsh /var/log | head -n 5
total 1.8M
-rw-r--r-- 1 root root 1.4M Sep  3 04:01 dpkg.log
-rw-r--r-- 1 root root 85K May 29 15:25 alternatives.log
-rw-r--r-- 1 root root 54K Sep  5 03:45 boot.log
-rw-r--r-- 1 root root 30K Sep  3 03:35 Xorg.0.log.old
```

9) Processer & services

1. Kør ping -c 10 8.8.8.8 og stop den med Ctrl+C.

```
(kali㉿kali)-[~]  
$ ping -c 10 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=22.5 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=18.0 ms  
^C  
— 8.8.8.8 ping statistics —  
2 packets transmitted, 2 received, 0% packet loss, time 1002ms  
rtt min/avg/max/mdev = 17.982/20.217/22.453/2.235 ms
```

2. Start sleep 120 i baggrunden og stop den igen.

```
(kali㉿kali)-[~]  
$ sleep 120 & kill %1  
[1] 31584  
[1] + terminated sleep 120
```

3. Tjek status for ssh-service.

```
(kali㉿kali)-[~]  
$ systemctl status ssh  
o ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)  
   Active: inactive (dead)  
     Docs: man:sshd(8)  
           man:sshd_config(5)
```

Aktive ssh:

```
(kali㉿kali)-[~]  
$ systemctl status ssh  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)  
   Active: active (running) since Sat 2025-10-11 12:00:00 CEST; 1min 1s ago  
     Docs: man:sshd(8)  
           man:sshd_config(5)
```


🔒 10) Kryptografi (basic): hash, kryptering, signatur

1. Lav en SHA-256 hash af .hidden_notes og gem den.

```
kali-linux-2025.2-vmware-... x
kali@kali: ~/kali-ovelses/fs/tmp
File Actions Edit View Help
(kali@kali)-[~]
$ cd ~/kali-ovelses/fs/tmp && sha256sum .hidden_notes > notes.sha256
(kali@kali)-[~/kali-ovelses/fs/tmp]
$ cd ~/kali-ovelses/fs/tmp && sha256sum -c notes.sha256
.hidden_notes: OK
```

2. Krypter .hidden_notes symmetrisk til en ny fil og dekrypter igen.

```
(kali@kali)-[~/kali-ovelses/fs/tmp]
$ cd ~/kali-ovelses/fs/tmp && gpg --symmetric --output notes.gpg .hidden_notes
[15519]@kali (gpg --symmetric --output no
Passphrase:
Enter passphrase
Password:
Confirm:
Save in password manager
Cancel OK
```

```
(kali@kali)-[~/kali-ovelses/fs/tmp]
$ cd ~/kali-ovelses/fs/tmp && gpg --decrypt --output notes.dec notes.gpg && diff -u .hidden_notes notes.dec || true
gpg: AES256.CFB encrypted data
gpg: encrypted with 1 passphrase
```

3. (Med nøgle) Signér .hidden_notes og verificér signaturen.

```
(kali@kali)-[~/kali-ovelses/fs/tmp]
$ gpg --quick-generate-key "Lab User" default default never

We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: revocation certificate stored as '/home/kali/.gnupg/openpgp-revocs.d/F2471D5E0D51BD740976E74592D2992EA65A7997.rev'
public and secret key created and signed.

pub  ed25519 2025-09-16 [SC]
      F2471D5E0D51BD740976E74592D2992EA65A7997
uid                               Lab User
sub  cv25519 2025-09-16 [E]

(kali@kali)-[~/kali-ovelses/fs/tmp]
$ cd ~/kali-ovelses/fs/tmp && gpg --output notes.sig --detach-sign .hidden_notes

(kali@kali)-[~/kali-ovelses/fs/tmp]
$ cd ~/kali-ovelses/fs/tmp && gpg --verify notes.sig .hidden_notes
gpg: Signature made Tue 16 Sep 2025 08:54:25 AM EDT
gpg:      using EDDSA key F2471D5E0D51BD740976E74592D2992EA65A7997
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: Good signature from "Lab User" [ultimate]
```

```
(kali㉿kali)-[~/kali-ovelser/fs/tmp]
$ (cd ~/kali-ovelser/fs/tmp && gpg --verify notes.sig .hidden_notes)

gpg: Signature made Tue 16 Sep 2025 08:54:25 AM EDT
gpg:          using EDDSA key F2471D5E0D51BD740976E74592D2992EA65A7997
gpg: Good signature from "Lab User" [ultimate]
```

```
(kali㉿kali)-[~/kali-ovelser/fs/tmp]
$ cat notes.dec
hej kali

(kali㉿kali)-[~/kali-ovelser/fs/tmp]
$ cat notes.gpg
♦
♦j♦BCd♦ K♦0♦k♦=♦n♦♦♦♦♦♦♦♦T♦q{♦♦vh-

(kali㉿kali)-[~/kali-ovelser/fs/tmp]
$ cat notes.sha256
09f062b24fd1c3bb7091b99180a7451e04599b484fe47e3ddb52d53c0333cc98 .hidden_notes
```

🔒 11) AI i shell

Undersøg applikationen shell-gpt: <https://pypi.org/project/shell-gpt/>

Undersøgt - Det er et fedt værktøj. Det kan effektivisere ens arbejde, hvis man har brug for hjælp til en forklaring eller generere bash-kommandoer.